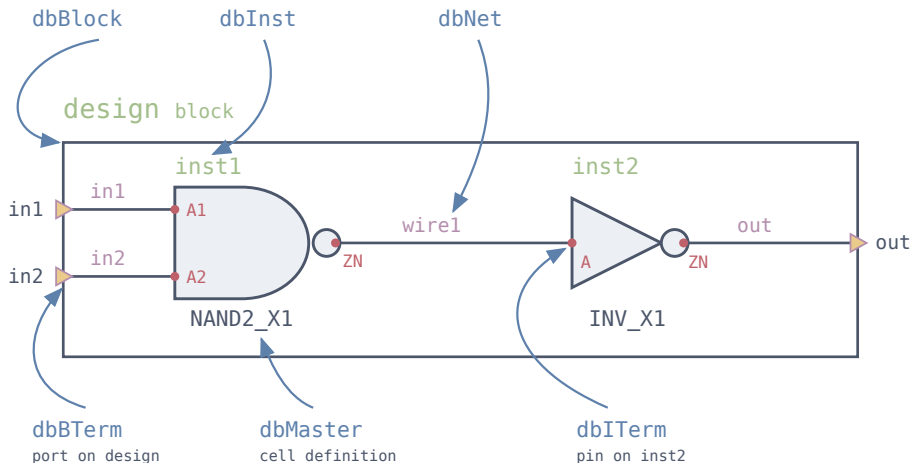


Ports (BTerms), cell pins (ITerms), wires (nets)



Lengths and areas across the layout views

View	Example	Meaning
LEF	SIZE 0.76 BY 1.4	Width and height in μm
DEF	UNITS DISTANCE MICRONS 2000	2000 database units = 1 μm
Tcl	detailed_placement -max_displacement 10	Movement limit of 10 μm
GUI	Cursor coordinates and ruler	Distances in μm

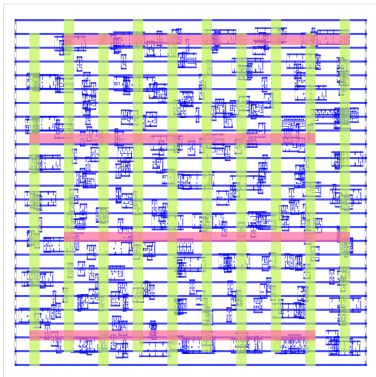
A cell that measures $0.76 \times 1.4 \mu\text{m}$ has an area of $1.064 \mu\text{m}^2$. Area uses square units;

coordinates and distances use length units.

Four ways to do global placement

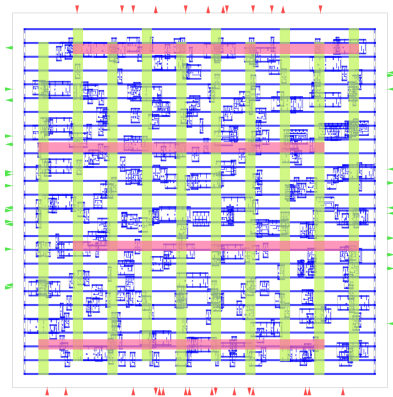
Mode	Pins	Needs
<code>global_placement</code>	Uses placed pins; <code>-skip_io</code> ignores them	Floorplan, cells, and rows
<code>-place_ios</code>	Co-optimizes movable pins and cells	Supported die and pin setup; run <code>place_pins</code> afterward
<code>-routability_driven</code>	Uses placed pins	Routing layers for congestion estimates; no prior routing or parasitic estimate
<code>-timing_driven</code>	Uses placed pins	Liberty, timing constraints, and wire RC; CTS is not required

Routing-aware placement setup



```
# Reserve 50% for later routing.  
# Keep placement routable.  
# Set signal and clock layer ranges.  
set_global_routing_layer_adjustment \  
    metal2-metal10 0.5  
set_routing_layers \  
    -signal metal2-metal10 \  
    -clock metal6-metal10
```

Final routability-driven placement



```
# Placement mode alternatives:
```

```
-place_ios
```

```
# Co-optimize pin placement with  
# cell placement.
```

```
-skip_io
```

```
# Place cells without considering  
# or needing pin locations.
```

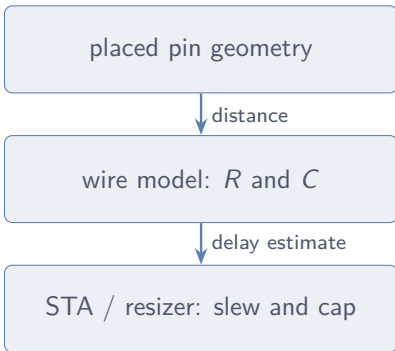
```
# Density is between 0 and 1.
```

```
# Default is 0.7.
```

```
# Final pass: use congestion in the  
# objective after pins are placed.
```

```
global_placement \  
-routability_driven \  
-density 0.80 \  
-pad_left 2 -pad_right 2
```

Estimate placement parasitics



placement-based parasitics
approximate future routed wires

```
# One set_layer_rc call per layer.  
# -layer takes one name, not a range.  
# Tcl foreach can reuse R/C values  
# across groups of layers.  
# E.g. M1-M4 and M6-M7, separately.  
# Override missing or unsuitable  
# LEF values only.  
# R: kohm/um; C: fF/um in Nangate45.  
set_layer_rc -layer metal3 \  
    -resistance 5e-3 \  
    -capacitance 5e-2  
# set_wire_rc chooses signal and  
# clock layers for wire estimates.  
set_wire_rc -signal -layer metal3  
set_wire_rc -clock -layer metal6  
# Use placed pin locations, not  
# detailed routed wires.  
estimate_parasitics -placement
```

How repair_design fixes violations

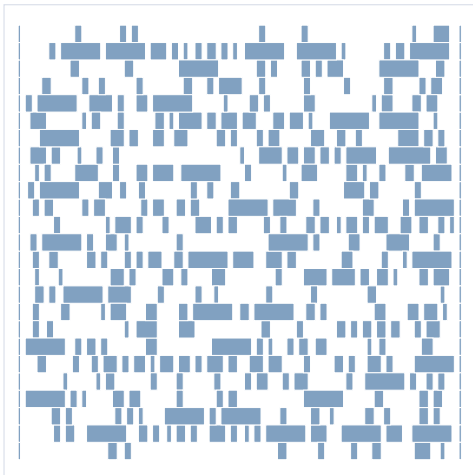
Violation	Problem	Typical fix
Slow transition	Signal edge exceeds the slew limit	Insert buffers or resize the driver
High fanout	One net drives too many pins	Build a buffer tree for the loads
Long wire	Wire RC adds too much delay	Insert buffers along the wire

Estimate with `-placement` before repair and again after CTS.
After global routing, re-estimate with `-global_routing`.

Placement engines

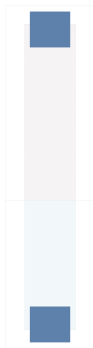
- ▶ Supports mixed cell heights (1x–4x)
- ▶ Supports fence regions and fragmented rows
- ▶ Supports placement padding and filler insertion
- ▶ Two legalization engines:
 - ▶ **Negotiation legalizer (default):** handles dense placement, but is slower; uses diamond search for remaining corner cases.
 - ▶ **Diamond search (legacy):** faster search from each cell's starting position. Enable with `-use_diamond_legalizer`.

Detail placement

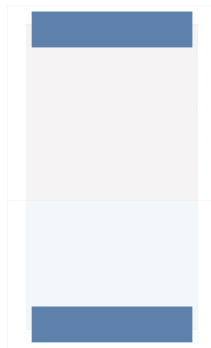


```
# Reserve space beside each cell.
set_placement_padding -global \
    -left 1 -right 1
# Move cells onto legal row sites.
# Resolve overlaps after repair.
# Cap movement to 10 um in each axis.
detailed_placement \
    -max_displacement 10
# Check for placement violations.
check_placement -verbose
```

Nangate45 filler cells in GDS



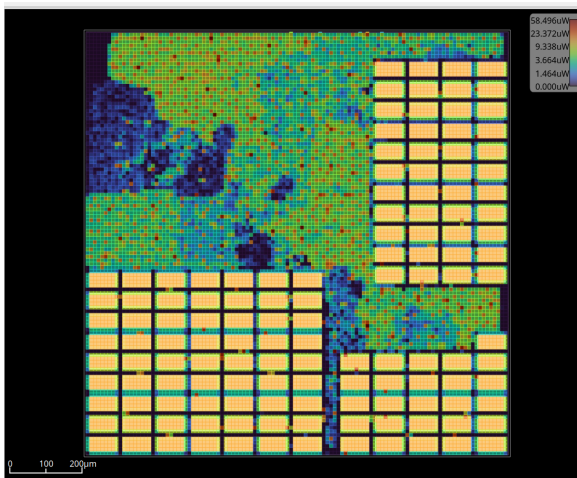
FILLCELL_X1: $0.19 \times 1.4 \mu\text{m}$



FILLCELL_X4: $0.76 \times 1.4 \mu\text{m}$

These cells fill row gaps and carry the VDD/VSS rails.

Heatmaps after placement



- **Power:** `report_power`;
Heat Maps > Power. Uses activity and estimated RC. CTS and buffers will change it.
- **Pin density:** Heat Maps > Pin.
- **Cell density:** Heat Maps > Placement.